Quantum State Discrimination with Bosonic Channels and Gaussian States

Course: PH3203 Advanced Quantum Mechanics
Instructor: Professor Sourin Das

A Term Paper Review of Chapters 1–3[1]

Based on the MIT Doctoral Thesis by Si Hui Tan (2010)

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Abstract

[ABSTRACT]

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1 Introduction

2 Continuous Variable Quantum Information Theory

3 State Discrimination Using Gaussian States

3.1 Distinguisability Of States

Although, the problem of distinguisability is defined for any number of states, let us start with the problem of distinguisability of two states, $\hat{\rho}_0$ and $\hat{\rho}_1$, with their corresponding equiprobable binary hypothesis H_0 and H_1 respectively. However, so far, we have only discussed more of the formalism, but nothing that could be used quantify the distinction between two states. In particular, we want something that can quantify some *distance* between the two states or some kind of method to quantify some *error* for distinction between two states given the measurement. For the notion of *distance*, we use *fidelity of states* and as discused in chapter 1, the *Helstrom's Bound* can be viewed as a measure of the error which gives the minimum probability of error and the corresponding POVM that gives the minimum measurement errors.

3.1.1 Helstrom's Optimal Discrimination

As discussed in chapter 1, Helstrom's Bounds corresponds to the generic optimization problem to find the POVM that gives the best measurement with minimum error probability. Given the two states, we consider a dichotomous POVM $\{\hat{E}_0, \hat{E}_1\}$, where $\hat{E}_0$ corresponds to the hypothesis $\hat{H}_0$ and $\hat{E}_1$ corresponds to the hypothesis $\hat{E}_1$ respectively.

References

[1] Si Hui Tan. "Quantum State Discrimination with Bosonic Channels and Gaussian States". In: *MIT Doctoral Thesis* (2010), pp. 0–70.